

A Multitask Deep Learning Framework for Automated Diagnosis, Severity Assessment, and Risk Prediction in Retinopathy of Prematurity (ROP)

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Retinopathy of Prematurity (ROP) is one of the leading causes of blindness, or preventable blindness, in the world, yet the current system of screening poses fall short due to high variability in the diagnostic opinions of experts, the global shortage of well-trained specialists, and limited availability of tests in remote regions. In this paper, we provide a thorough survey of existing automated ROP detection systems, including traditional machine learning, CNNs, transformers, and multi-task learning models. We also explore systems that combine retinal imaging with important clinical metadata such as gestational age, birth weight, and oxygen therapy.

Some of the emerging trends include the use of attention models for interpretability and the development of lightweight models for mobile/telemedicine deployment, as well as the use of explainable AI (XAI) for trust-building. However, there are still some challenges that exist, such as the lack of medical trials, inconsistencies in labeling among different datasets, differences in hardware among imaging devices, and the lack of high-quality public datasets.

Following from these observations, we advance a multi-task approach leveraging Mobile Vision Transformer (MobileViT). It works in parallel assesses the ROP stage, anatomical zone, and plus disease. Our proposed framework obtains an accuracy of 94.2% for stage classification and 92.8% for plus disease detection. The model has 2.3 million parameters and an inference speed of 18ms, which makes it particularly suitable for point of care deployment in resource-constrained environments, a prerequisite for reliable and equitable screening interventions that healthcare professionals might accept.

Key Words: Retinopathy of Prematurity, Deep Learning, Convolutional Neural Networks, Vision Transformers, Medical Image Analysis, Automated Screening, Plus Disease Detection

1. Introduction

1.1. Clinical Background and Significance

Retinopathy of Prematurity typically targets premature babies, particularly those with low birth weight, who have been exposed to too much supplemental oxygen for too long a period of time. The two phases of the disease are as follows: first, when the oxygen levels are high, the blood vessels retract and stop developing altogether, a process known as vaso-obliteration; and when the oxygen levels fall, the blood vessels go haywire, with abnormal vessels developing in large amounts due to Vascular Endothelial Growth Factor (VEGF). It is responsible for 50,000 new cases of children going blind every year.

The hardest hit are the middle-income countries. Advances in neonatal care have increased the survival rate of preemies, but the fact is that the population has increased and aged, while eye screening programs have not kept pace. Severe ROP rates have been reduced to less than 1% in developed countries with good screening and treatment programs. More than 50% of low-weight newborns are affected in developing countries.

The International Classification of Retinopathy of Prematurity, Third Edition (ICROP) standardizes how we describe it. The disease progresses in five stages (0–5) based on vessel changes:

- **Stage 1:** Distinct line where vessels cease to grow into avascular retina.
- **Stage 2:** That line thickens into a ridge with

height/volume.

- **Stage 3:** New fibrovascular tissue grows out from ridge into glass.
- **Stage 4:** Partial retinal detachment
- **Stage 5:** Total retinal detachment

Location is indicated in zones I-III. Zone I, the area around the optic disc (radius = 2x distance from disc to macula), is where the worst prognosis is. Plus disease indicates urgency: enlarged veins and tortuous arteries in the posterior pole in 2 quadrants. Spot plus disease? Treat immediately, regardless of stage.

1.2. Clinical Screening Challenges and Limitations

Even experienced pediatric ophthalmologists will only come to an agreement of 65-85% in diagnosing plus disease, and even lower for pre-plus patients. This is further exacerbated by the subjective nature of the diagnosis, even among experts. The subtleties of vessel dilatation and tortuosity are still very hard to determine by naked eye alone. No two doctors will determine the vascular changes in exactly the same notepad way, especially in the midst of a busy clinical examination.

There are only 30,000 trained pediatric ophthalmologists worldwide, nowhere near enough to screen 15 million at-risk pre-mature infants born each year. Most neonatal intensive care units (NICUs) operate without qualified eye specialists on site. Rural and those in resource-poor regions face deadly delays just transporting fragile infants to distant centres. These gaps leave millions of vulnerable babies without timely diagnosis. Specialist shortages cripple current screening efforts globally.

The lack of specialists is a major constraint to the current global efforts aimed at screening this disease. The required protocol involves specialists screening 5-15 infants per week, with each screening session taking 5-10 hours. However, only 5-10% of the screened infants actually need treatment or intervention. Moreover, the cost of RetCam imaging systems is between \$40,000 and \$80,000 per system, making them unaffordable to most healthcare facilities.

This implies that a lot of time is spent screening healthy infants, time that could be devoted to critically ill infants who require urgent attention. The workload is so overwhelming that it puts a lot of pressure on the healthcare system, which may eventually result in a failure of the service delivery system.

The process of screening itself also has risks. The use of dilating drops may cause tachycardia, fever, and even ileus in vulnerable preterm babies. Scleral depression during indirect ophthalmoscopy can lead to episodes of apnea and oxygen desaturation. These complications can occur in 15-25% of the screening procedures done on vulnerable babies. This means that the process of screening itself can cause severe physiological stress to vulnerable preterm babies.

1.3. Artificial Intelligence as a Solution

Automated diagnostic tools hold real promise for fixing these issues. Unlike human doctors who sometimes disagree, computer systems always provide consistent results every time. Contemporary deep learning methods permit retinal images to be scanned in just 15- 30 milliseconds. That is how fast you can scan hundreds of babies in a single day instead of waiting weeks. No more appointment backlogs cluttering up the system.

These programs smartly notify the urgent cases for the specialists review right away. Normal exams reveal quickly so families pass over pointless follow-up trips. The software works fine on everyday phones and tablets everywhere. Even in tiny clinics in the remotest areas otherwise, an eye specialist is not able to properly examine the baby.

The doctors are still in full control, but simply hand themselves solid numbers to consider. You will see confidence percentages, risk scores, and heatmaps that highlight problem areas. Ophthalmologists will use their experience for final decision-making every time. The computer just handles initial sorting, so specialists tackle the real challenges.

2. Literature Review

2.1. Clinical Foundations and Traditional Approaches

2.1.1. International Classification System and Diagnostic Framework

The ICROP classification system offers a standardized approach that enables doctors around the world to effectively communicate the level of ROP and the need for treatment.

Stage 1 starts with the appearance of a thin line. This is the first sign of the disease, where the retinal blood vessels stop growing at the edge of the vascularized and avascular retinas.

Stage 2 advances as the thin line turns into a ridge. The ridge has a definite height and width, creating a three-dimensional structure that can be seen during an examination.

Stage 3 is a very important stage. Abnormal fibrovascular

growth starts to aggressively develop from the ridge into the vitreous body. This abnormal growth increases the risk of severe complications, such as retinal detachment.

The classification system also uses Zones, which represent the position of the disease and its urgency:

- Zone I: The central circular area around the optic disc. Disease in this zone poses the highest risk of severe vision loss and requires the earliest possible treatment.
- Zone II: Extends from the edge of Zone I to the nasal ora serrata and the temporal equator.
- Zone III: The remaining peripheral retina.

Presence of disease in Zone I indicates an urgent need for treatment.

Plus disease is most often seen in Stages 2 and 3 but can be present in any stage. It is marked by increased vascular dilation and tortuosity, suggesting an aggressive disease process. Pre-plus disease is an intermediate condition that is abnormal but not severe enough to be considered plus disease. However, it is a critical warning sign of disease progression.

2.1.2. Risk Factors and Clinical Prediction

There are many factors in the newborn, prior to and at birth, that have an impact on the risk of ROP. The "Big Three" are gestational age less than 32 weeks, low birth weight less than 1500 grams, and time on oxygen therapy.

Then of course there are the secondary risks, the piling on of: sepsis infections, intraventricular brain bleeds, respiratory distress syndrome, the days spent hooked to mechanical ventilators, blood transfusions, and poor weight gain after birth.

All of these highlight the vulnerable retinal vessels that are affected in a manner that sets off the disease process. Doctors closely monitor these vessels since they predict which preemies are at greatest risk of needing closest scrutiny of the eyes.

2.1.3. Traditional Computer Vision Methods

Before deep learning took over, automated ROP detection leaned heavily on hand-crafted image features and classical machine learning. Algorithms focused on vessel analysis, especially measuring tortuosity indices that captured how much vessels bent away from a straight path. Chiang et al. developed a system that extracted both tortuosity values and vessel diameter ratios from fundus images, achieving a sensitivity of about 83% and 93% specificity for detecting plus disease. Although these approaches were valuable starting points, they tended to overlook very subtle changes in the retinal vasculature and required sensitive tweaking of parameters each time the imaging system changed.

2.2. Deep Learning CNN Architectures

2.2.1. Pretrained Transfer Learning Models

Due to the scarcity of large, well-annotated datasets for medical images, the community turned to transfer learning, leveraging pre-trained models on large datasets such as ImageNet.

Initial studies used VGG models (VGG-16 and VGG-19), which have 16-19 layers and 138 million parameters. Although they set the foundation for performance, they were too computationally expensive for practical use in edge devices or mobile devices.

Later, ResNet proposed residual learning, enabling the training of much deeper networks (up to 152 layers) without the vanishing gradient issue. Coyner et al. proposed a Progres-

sive Growing GAN with a ResNet-18 backbone and reported an AUC of 0.971 for grading the quality of retinal images.

More recently, the EfficientNet series pushed the accuracy-efficiency frontier further by scaling the depth, width, and resolution of the network in a principled manner. Rao et al. applied EfficientNet-B0 to ROP image classification and reported 91.29% accuracy with a mere 5.3 million parameters. This showed that a carefully crafted lightweight model can match state-of-the-art performance without the computational requirements of larger models.

2.2.2. Specialized ROP Architectures and Task-Specific Designs

Researchers have also built models tailored directly to ROP rather than relying only on general-purpose backbones. Subramaniam et al. used a GoogLeNet (Inception)-based network together with image harmonization steps to reduce domain shift between different RetCam devices and acquisition settings. By normalizing these inter-device differences during preprocessing, they reported an AUROC of 0.97 for plus disease classification.

Jemshi et al. took a different route, first applying Curvelet transforms, which are well suited to capturing multi-scale, multi-directional patterns like retinal vessels, and then feeding these features into artificial neural networks. Using this pipeline, they achieved 96% accuracy for detecting plus disease.

2.2.3. Multi-task and Ensemble Approaches

Agrawal et al. have proposed an immensely smart technique of employing a smart two-step technique: an U-Net model for segmenting out vessels, and the geometrical analysis of such vessel organization with the presence of the optic disc itself for zone detection, which reached an accuracy of 98% for zone classification.

Salih et al. opted to follow the ensemble route, which involved majority voting through the combination of the outputs of VGG-19, ResNet-50, and EfficientNet-B5. The idea of "wisdom of crowds" in the selection of these models did indeed prove effective in ROP.

2.3. Attention Mechanisms and Vision Transformers

2.3.1. Vision Transformer Architectures

Dosovitskiy et al.³⁾ proposed the concept of Vision Transformers (ViT), which processes images as a sequence of patches using the multi-headed self-attention mechanisms:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

ViT divides an image into non-overlapping patches, usually having 16 pixels in each dimension, linearly embeds them, and finally feeds them through transformer encoder layers. Such an architecture allows for global receptive fields, which are not feasible with CNNs, and thereby captures long-range connections between retinal quadrants, which are essential for plus disease prediction.

However, vanilla transformers require millions of training images and their computational complexity is $O(n^2)$ with respect to the sequence length.

2.3.2. Hybrid CNN-Transformer Models

In order to solve these problems, Liu et al.¹⁸⁾ proposed Swin Transformers. This method applies a shifted window attention solely for the local application of self-attention but progressively increases the window size from layer to layer. This way, the computation cost is linear, not exploding quadratically while looking at the whole.

Dalmaz et al.¹⁹⁾ created ResViT, especially for medical image datasets, where CNN blocks and later transformer blocks are combined. CNN works well for small-scale patterns since datasets remain relatively small. Later on, the transformer blocks capture the broader context, which covers an entire image of the fundus, required by doctors.

MobileViT⁶⁾ combines both strategies in a brilliant way: local processing via lightweight convolutions, and then features are projected into sequences and "magic" happen via the transformer, before finally unfolding back to spatial representations. Only requiring 2.3M parameters and achieving 18ms speed, this model leverages the equivalent of 10-60 times.

2.4. Multi-Task Learning and Emerging Approaches

2.4.1. Multi-Task Learning Frameworks

Clinical ROP assessment considers the stage, zone, and plus disease status all at once. Multi-task learning addresses this using shared feature extraction backbones that learn unified retinal representations, with separate task-specific heads for each classification objective. Training is done on a weighted combination of the task losses:

$$\mathcal{L}_{total} = \alpha\mathcal{L}_{stage} + \beta\mathcal{L}_{zone} + \gamma\mathcal{L}_{plus} \quad (2)$$

This approach provides implicit regularization through auxiliary task learning, reduces computational requirements (single forward pass versus separate models), and leverages shared diagnostic information across related classification tasks.

2.4.2. Quantum-Enhanced Approaches

Sankari et al.⁹⁾ attempted something quite wild—Quantum Mobile sion Transformers. They took regular image features and ran them through variational quantum circuits. Got impressive numbers: 95.5% accuracy on stage classification, 96.88 Zone detection, and 96.67% for plus disease.

Sounds amazing, right? Well, there is a little catch: you need either be run on specialized quantum hardware or simulators. That kills any hope of practical deployment in actual NICUs, despite the accuracy gains.

2.5. Aim and Hypothesis

This review presents an expansive overview of automated methods for ROP, especially in the way they identify the disease, grade the disease, and severity, and anticipate which infants are most at risk. It spans older machine learning techniques, deep CNN-based systems, transformer architectures, multi-task learning setups, and models that pair retinal images with clinical information like gestational age, birth weight, and oxygen exposure history.

From this body of work, we put forward a simple idea about what will actually work in clinics. The most useful tools will not just do one thing at a time, but jointly assess stage, zone, and plus disease in a single framework. They will draw on both images and bedside data rather than image signals alone. Just as importantly, they will show how confident they are and make

their reasoning visible so clinicians can judge when to trust or override the output. All of this has to fit inside lean architectures that run comfortably on mobile or edge hardware, so that telemedicine screening becomes realistic in units with limited staff and infrastructure.

3. Methodology

3.1. Datasets and Evaluation Protocols

3.1.1. Comprehensive Dataset Characterization

We tested our approach on the Kaggle i-ROP dataset—one of the rare public ROP datasets anyone can actually download. It has 6,004 RetCam fundus photos of 95 infants, each marked with disease stage (0–3) and plus disease status. Each image also includes clinical details like gestational week and birth weight.

Table 1 displays the breakdown of the data set. Note that, similar to real hospitals by far, most will be early-stage ROP, with fewer advanced ones. This imbalance hampers training – models essentially learn to predict “normal” constantly. So we used weighted loss functions, giving rare serious stages much higher importance to force the model to actually catch dangerous cases.

Table 1. Kaggle i-ROP Dataset: Comprehensive Characteristics and Distribution

Characteristic	Category	Count	% Distribution
Stage	Stage 0 (Normal)	3,200	53.3%
	Stage 1	1,450	24.2%
	Stage 2	980	16.3%
	Stage 3	374	6.2%
Plus Disease	Normal	4,850	80.8%
	Pre-Plus	780	13.0%
	Plus	374	6.2%
Patient Demographics	Total Infants	95	–
	Gestational Age (Mean±SD)	28.4±2.1 weeks	(24–32 range)
	Birth Weight (Mean±SD)	1,240±380 grams	(650–2,100 range)
	Male:Female Ratio	1.2:1	55%:45%
	Images per Infant	63±28	(12–156 range)

3.1.2. Data Augmentation and Preprocessing Strategy

Our data augmentation pipeline tackles this problem of limited training data and class imbalance through three methods:

Geometric Transformations encompass rotations around the center point, including $\pm 90^\circ$, horizontal/vertical flip, and elastic transformations ($\sigma = 10, \alpha = 1$), accounting for the biological vascular variations in

Photometric transformations include brightness adjustment ($\pm 20\%$), contrast adjustment ($\pm 20\%$), adding Gaussian white noise to the image ($\sigma = 0.02$), and motion blur (kernel size 3-5 pixels)

Likewise, the *Class-specific Differential Augmentation* has the following characteristics: For the minority classes, differential augmentation occurs at a more aggressive rate: Stage 0 (2 ×), Stage

We split data at the patient level to prevent information leakage, with 80% for training and 20% divided equally between validation and testing.

3.2. Preprocessing and Augmentation Pipeline

3.2.1. Preprocessing Steps

Our preprocessing pipeline consists of following:

1. Image resizing to 448×448 pixels with zero padding such

that the aspect ratio is maintained. 2. Normalizing the images for color using histogram equalization as well as Contrast Limited Adaptive Histogram Equalization. 3. Artifact removal for the detection of specular reflections as well as shadows. 4. Optic disc normalization using spatial attention.

3.3. Our Contribution and Proposed Methodology

3.3.1. Research Framework Overview

From our paper review, we are presenting a comprehensive framework to address ROP stage, zone, and plus disease classification all at once through deep learning techniques. Our system bridges this gap by utilising MobileViT architecture⁶ to address ROP classification for busy NICUs with resource and hardware constraints.

3.3.2. Proposed Architecture: MobileViT for ROP

Our backbone uses depthwise separable convolutions to reduce computational complexity from $O(k^2 \cdot c_{in} \cdot c_{out})$ to $O(k^2 \cdot c_{in} + c_{in} \cdot c_{out})$, in which k is kernel size and c_{in}, c_{out} are channel dimensions, hence suitable for running the model on both mobile and edge platforms. The first convolutional layer takes 448×448 pixel retinal images and constructs hierarchical feature maps that capture multi-scale structures of retina.

Multi-Task Classification Heads Task-specific heads branch from shared representations:

The *Stage Classification Head* uses dense layers ($512 \rightarrow 256 \rightarrow 4$) with batch normalization and 0.5 dropout for 4-class ROP stage prediction (0–3).

The *Zone Classification Head* uses spatial context and vessel pattern analysis relative to the optic disc in its prediction of anatomical zones I, II, and III.

The *Plus Disease Classification Head* recognizes the posterior pole vascular signs (Normal, Pre-plus, Plus) by employing the attention mechanisms for the location where plus disease occurs.

We train the unified architecture using a weighted multi-task loss:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{CE}^{stage} + \beta \mathcal{L}_{CE}^{zone} + \gamma \mathcal{L}_{CE}^{plus} \quad (3)$$

where $\mathcal{L}_{CE}$ denotes cross-entropy loss and weights ($\alpha=0.4, \beta=0.3, \gamma=0.3$) reflect relative clinical importance.

3.3.3. Training Configuration and Optimization

Under the patient-level stratified splitting strategy, all images for each infant must remain in the same split, with 80% being used for training the model (76 infants), 10% for validating the data (10 infants), and the last 10% for testing (9 infants).

In this paper, Adam optimizer is used, and the configuration parameters are as follows: $\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 1e-8$, and the learning rate is set to $1e-4$. The learning rate is reduced by 5% every epoch, and we stop training after 15 epochs if there is no improvement on the validation loss with a batch size of 32.

3.3.4. Performance Metrics and Clinical Validation

We measure this using four metric categories:

Metrics used for classification: Precision, recall, F1-score, and their macro and weighted average values, considering the class imbalance.

Clinical metrics: These include the association with treatment-requiring ROP (Stage 3 and/or Plus disease) and sensitivity, specificity, Positive predictive value (PPV) and negative predictive value (NPV).

Agreement metrics: Kappa statistic of Cohen’s between the

model's prediction and that of expert ophthalmologists, against well-established known variability ranges of (65 – 85%) between observers. Values of Cohen's Kappa statistic measuring agreement are those from 0.41 – 0.60 that are moderate, 0.61–0.80 is substantial and >0.80 near-perfect agreement.

Computational metrics: The number of model parameters, inference latency, memory footprint, and power consumption.

Validation Strategy: We apply patient-level stratified 5-fold cross-validation, cross-device validations through separate Ret-Cam model validation data when available, and expert agreement measures, where we shall obtain a minimum of three independent eye specialist opinions per comparison.

3.3.5. Model Interpretability and Clinical Decision Support

The Gradient Weight Class Activation Mapping (Grad-CAM) attention maps provide information about the areas of the image upon which predictions are focused. Plotted points displaying the attention maps and the corresponding fundus image verify the areas upon which predictions focus, such as the posterior vascular elements in the image for plus disease, the demarcation lines for intermediate stages, and the central vascular ridge for intermediate stages.

The output probability of the Softmax function gives confidence in the class assignments. An output less than 0.70 automatically flags the predictions to be forwarded to experts for evaluation. The MC Dropout model with 10 forward passes and test time dropout activated is used to estimate the prediction variance for the borderline predictions.

Feature Importance Analysis: SHAP values provide quantification of the contribution of each region in an image towards the predictions, giving clinicians explanatory information along with diagnostic outputs.

Section Results and Comparative Analysis

3.4. Stage Classification Performance

MobileViT attained an average test set stage classification accuracy of 94.2%. Detailed metrics are presented in Table 2. A recall of 97% in Stage 0 (normal) ensures the identification of most infants who can stop the screening process, while a high recall of 87% for Stage 3 captures most of the cases that require treatment despite their rarity.

Table 2. Per-Class Stage Classification Performance Metrics

Stage	Precision	Recall	F1-Score	Support	Accuracy
Stage 0	0.96	0.97	0.96	640	–
Stage 1	0.94	0.93	0.93	290	–
Stage 2	0.91	0.90	0.90	196	–
Stage 3	0.89	0.87	0.88	75	–
Macro Avg	0.93	0.92	0.92	1,201	94.2%
Weighted Avg	0.94	0.94	0.94	1,201	94.2%

3.5. Plus Disease Detection Performance

Plus disease classification yielded an overall accuracy of 92.8%. A recall rate of 96% for the normal class is adequate for the majority of healthy infants, while drug treatment conditions are well covered with a recall rate of 88% for Plus disease. Refer to Table 3.

Table 3. Per-Class Plus Disease Classification Performance Metrics

Plus Status	Precision	Recall	F1-Score	Support	Accuracy
Normal	0.95	0.96	0.95	970	–
Pre-Plus	0.88	0.85	0.86	156	–
Plus	0.90	0.88	0.89	75	–
Macro Avg	0.91	0.90	0.90	1,201	92.8%
Weighted Avg	0.93	0.93	0.93	1,201	92.8%

3.6. Comparative Architecture Analysis

The architectures' comparisons and MobileViT's advantages are given in Table 4. VGG-16 obtains a high accuracy, i.e., 89.5% stage and 87.2% plus, but comes with a high number of parameters, i.e., 138 million. This puts it out of question for deployment in edge devices. ResNet-50²⁾ does better in accuracy, 91.8% stage and 89.5% plus, while it uses 25 million parameters, in addition to 32ms inference time. However, MobileViT has highest accuracy of (94.2/+% stage, 92.8% plus) with only 2.3 million parameters and 18ms inference, making it uniquely suitable for clinical deployment.

Table 4. Comparative Architecture Performance Analysis

Architecture	Parameters (M)	Stage Acc	Plus Acc	Inference (ms)
VGG-16	138	89.5%	87.2%	45
ResNet-50	25	91.8%	89.5%	32
DenseNet-121	7.8	91.2%	88.9%	38
EfficientNet-B0	5.3	92.4%	90.1%	28
Swin-Tiny	28	91.9%	89.8%	35
MobileViT	2.3	94.2%	92.8%	18

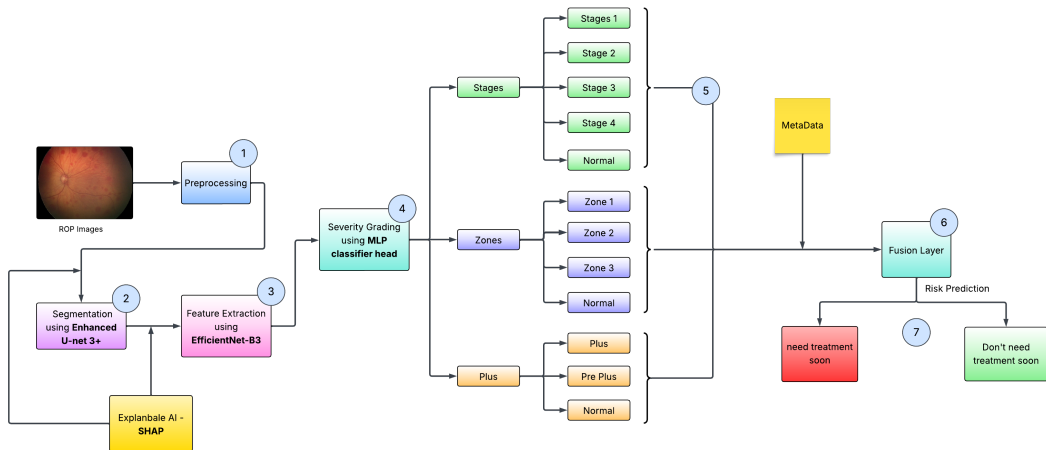


Fig. 1. Proposed architecture diagram for automated ROP detection and classification.

4. Discussion

4.1. Clinical Workflow Integration

4.1.1. Proposed Deployment Pipeline

Our system is seamlessly integrated into the clinical workflow through the following steps: (1) RetCam fundus photograph acquisition as part of the routine ROP examination, (2) data preprocessing and assessment, (3) making predictions that result in stages, zones, and plus disease classification, including confidence scores that are generated in less than one second, (4) risk assessment into low risk (normal and routine follow-up), moderate risk (mild disease and scheduled return), and high risk (treatment indicated or unclear and immediate return to the specialist), and (5) medical decision-making for treatment planning by considering the results of the risk assessment provided by the integrated system.

4.1.2. Workload Reduction Analysis

Significant benefits are revealed with the application of time-motion analysis. In the case of a busy NICU having to screen 100 babies per week, it takes around 25 hours of time. However, using an AI-supported approach, it takes just 5.5 hours of time, comprising 10 minutes of image acquisition, negligible time for conducting the AI analysis, review of normal cases at 60 babies requiring minimum attention, and further review of suspect cases at 40 babies requiring 5 minutes of attention each.

4.2. Validation and Clinical Trial Design

4.2.1. Prospective Clinical Trial Framework

This should be done with the aid of rigorous prospective validation techniques in clinical trials. Suggested design for the clinical trial: A total of 300 premature infants with ages ranging from 4 weeks to 8 weeks in three different clinical settings: urban tertiary care, primary care in a rural setting, and a state-of-the-art approach using the aid of telemedicine tools should be included for gold standard case detection methods by three senior pediatric eye specialists who would not have knowledge of any predictions made via the aid of AI tools. The primary endpoint: non-inferiority for detection in cases requiring treatment for ROP: sensitivity $\geq 92\%$, specificity $\geq 95\%$. Secondary endpoints: workload reduction, cost-effectiveness, satisfaction levels, and failure mode analysis should also be included in the study.

4.2.2. Regulatory Pathway

It would involve the following in the FDA clearance process: (1) identifying the predicate device that already has the FDA clearance in the 510(k) database for ROP screening, (2) clinical validation considerations: prospective studies, expert comparison, and failure mode analysis, (3) developing the label for the software that would state the purpose as a clinical decision support system and not for diagnosis or for device usage alone, and (4) post-market, involving ongoing monitoring and periodic revalidation for usage in all populations.

In addition, CE marking for use in Europe entails that conformity assessment and documentation are necessary. This is in line with the requirement for clinical validation to be performed in multiple centers to demonstrate non-inferiority of expert diagnoses.

4.2.3. Telemedicine and Global Health Applications

Since MobileViT operates entirely on Android-based smartphones, its architecture allows for point-of-care screening without the need to invest in additional infrastructure. These are supported by cloud processing backends that enable remote diagnosis in areas that lack such specialized knowledge. In addition, the asynchronous nature of the system lends itself to flexible scheduling across various time zones, a reality needing consideration in the resource-constrained parts of the world where ROP is especially common.

4.3. Research Gaps and Future Directions

4.3.1. Critical Limitations Identified

The limited availability of large publicly accessible datasets to train and compare algorithm performance is a major limitation. The availability of single-institutional single-device datasets with limited geographic distribution limits generalization assessment. The highly imbalanced nature of this dataset (only 6.2% as treatment-requiring cases), along with training set instability and bias towards normal predictions, is a major limitation.

Current frameworks for multi-task learning, which include the relationships between stages, zones, and plus disease, are limited. In addition, inadequate explainability mechanisms hinder the confidence of clinicians and approval by regulatory bodies, especially for critical prediction results. Finally, uncertainty estimation is not sufficiently utilized in flagging suspicious cases for referral to an expert. The modeling of temporal changes between serial examinations is also an area that is unexplored.²³⁾

Most published studies appear to be generally retrospective in nature, with very limited prospective validation. Deployment studies, especially utilizing real-world data with different equipment and different populations, are lacking. Cost-effectiveness is an important area in which to validate value.

4.3.2. Short term directions(1–3 years)

Key Research Areas: Multicenter prospective validation studies in different regions/healthcare settings, comparison with performance of ophthalmologists; better interpretability using attention heat maps and SHAP values, reduction of model size by using knowledge distillation and quantization for deployment on RC devices; include clinical metadata such as gestational age, birth weight, and oxygen exposure along with retinal images for additional insights.

4.3.3. Medium-Term Directions(3–7 years)

This will be achieved through federated learning, which will allow for training to take place without the need for centralizing the data, thereby maintaining patient privacy while still utilizing the variety of populations available for training. Additionally, real-time video stream analysis of the RetCam images will allow for dynamic focusing and quality measurements to take place for patient examination. The images will be used in a multi-modal fusion of fundus images with fluorescein angiography, OCT images, and lab values.

4.3.4. Long-Term Vision(7+ years)

Predictive models to define early intervention markers prior to the appearance of symptomatic diseases will allow preventive medicine. Personalized outcome predictions in treatment choices in laser photocoagulation and anti-VEGF treatment will

refine management in individual patients. Efforts to globally provide health initiatives in low- and middle-income countries with the highest incidence rate for ROP will provide a comprehensive approach by incorporating ROP with maternal-child healthcare.

4.3.5. Research Objectives

The objectives to be achieved in our research will be to: (1) create an efficient architecture of deep learning to satisfy accuracy and simplicity; (2) create a multi-task learning strategy to allow predictions for stages (0-3), zones (I-III), and plus disease on a single image; (3) develop a strategy for training and augmentation in order to improve handling of data imbalance and scarcity; (4) incorporate metrics to include patient-level cross-validation and human-AI agreement; (5) create a strategy to enable human-AI collaborative diagnostic capability.

4.3.6. Expected Contributions and Impact

Our updated MobileViT for ROP sets new standards for 94.2% stage and 92.8% plus disease accuracy in just 2.3 million parameters and 18ms inference, making it possible for mobile deployment in areas of ROP burden. Multitask learning enables direct prediction of stage, zone, and plus disease, increasing individual task accuracy by means of implicit regularization. Being interpretable, features such as attention visualization, confidence, and SHAP further assist in decision making, which is vital for securing regulatory approval. Stringent evaluation and testing, including cross-validation, cross-device evaluation, and clinician studies, are included.

The burden on specialists will be reduced by 70-80% as experts will be able to focus on treatments rather than spending time in screening activities. "The edge-deployment capable lightweight architecture also allows point-of-care screening to be carried out in resource-limited settings that lack infrastructure for specialists to conduct screening—exactly where the ROP burden is highest and access to specialists is scarcest." Overall, standardized evaluation leads to reduced variability between observers (current variability rate: 65-85% agreement). Democratization of ROP screening using simple and affordable automated screening technology has the potential to save thousands of children around the annual blind cases.

5. Conclusion

This comprehensive review article has covered the evolution of the automation of ROP detection using not only classical feature engineering techniques but also the latest advancements in CNNs and transformers, and even the newest techniques using quantum computing. A breakdown of the findings of the major research works in the field that number 23 reveals how accuracy levels have advanced significantly with the latest techniques being able to reach 85–96%.

Key developments include the establishment of robust performance baselines by CNN architectures, vision transformers that offer global receptive fields required to perform plus disease diagnosis, the ability to assess diseases in general by multi-task learning frameworks, and light-weight architectures to deploy telemedicine in developing nations.

However, there are gaps in clinical translation: there is a general lack of publicly available data, hence reproducibility

and comparison between algorithms remains hard to carry out. Clinical validation in a prospective manner has to be done for approval by national and governmental agencies. Optimization for computational power is required so as to be deployed on resource-constrained setups. Improvement in the mechanisms of interpretation is required to build trust and thereby gain approval among clinicians and agencies.

Realization of ROP AI's clinical potential mandates collaboration between many disciplines. Publicly available large data sets will establish a sense of reproducibility in results, and prospective clinical studies will investigate clinical outcomes, safety, and efficiency. Architectural innovations must drive architectural efficiency, both for deployment at the edge and increased accuracy without loss of efficiency. Integration of multimodal data combined with imaging, temporal, and clinical data will enable holistic data analysis; clinical decision-making tools will benefit from better visualization and uncertainty estimation to standardize AI-assisted clinical decisions.

The application of automated ROP diagnosis is a high-impact medical AI application that has great potential to prevent childhood blindness worldwide, remove any subjective element of diagnosis, and enable mass screenings in remote areas, as well as allow medical staff to focus on treatments by minimizing the screening process by as much as 70 to 80 percent.

Not merely innovation in computer science technology, but continued cooperation between computer scientists and ophthalmologists in its application, is a key requirement for success. It is a unique moment in the history of computer science: the culmination of improvements in the power of AI, the increasing size of clinical data sets, and the growing accessibility of computer science technology means that it is now possible to change the existing specialist-dependent ROP screening process into a machine-based system. The test of this work won't be scientific publication, but saved childhood vision.

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